

Conservativeness of the CUEP Energy Bound in IBR-Penetrated Networks: Proof, Monotonicity, and Application to the SAMBPS Protection Framework

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Abstract—The controlling unstable equilibrium point (CUEP) method provides an energy-based conservative lower bound on the critical clearing time (CCT) for transient stability assessment in synchronous machine networks. Whether this conservativeness is preserved when inverter-based resources (IBRs) alter the post-fault swing dynamics is an open question with direct implications for model-based differential protection. This paper proves three results. *First* (Theorem 1), the CUEP energy V_{cuep} is a strict lower bound on the true critical energy V_{cr} for any IBR penetration $k_{\text{ibr}} \in [0, k_{\text{ibr}}^{\text{max}}]$, provided the IBR current injection is bounded and the post-fault system has a unique stable equilibrium point (SEP). *Second* (Theorem 2), the conservatism gap $\Delta V = V_{\text{cr}} - V_{\text{cuep}}$ is monotonically non-decreasing in k_{ibr} : higher IBR penetration increases the gap, making the bound more conservative. *Third* (Proposition 1, the SAMBPS C13 result), for the SAMBPS protection framework, using V_{cuep} as the energy threshold for the trip/restrain decision yields zero false trips for all $k_{\text{ibr}} \leq 0.55$, at the cost of a CCT margin of 12–28 ms relative to the true CCT. Simulations on the IEEE 39-bus test system with a mixed IBR fleet (penetration 0–62%) confirm all three results and quantify the CCT conservatism as a function of k_{ibr} and fault location.

Index Terms—CUEP method, transient stability, inverter-based resources, energy function, critical clearing time, conservative bound, SAMBPS, model-based protection

I. INTRODUCTION

The controlling unstable equilibrium point (CUEP) method [1]–[3] is the central analytical tool for direct (non-simulation) transient stability assessment. Given a post-fault energy function $V(\delta, \omega)$ and the fault-on trajectory $(\delta(t), \omega(t))$, the method identifies the CUEP — the unstable equilibrium point whose stable manifold is closest to the fault-on trajectory — and uses the CUEP energy $V_{\text{cuep}} = V(\delta_u, 0)$ as the critical energy threshold. If the fault-cleared energy $V_0 = V(\delta_0, \omega_0)$ satisfies $V_0 < V_{\text{cuep}}$, the post-fault trajectory is predicted to remain within the stability region.

The method has two well-known properties for conventional synchronous machine (SG) networks:

- (i) *Conservativeness*: $V_{\text{cuep}} \leq V_{\text{cr}}$, where V_{cr} is the true critical energy (smallest energy level that contains an

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This work constitutes Contribution C13 of the SAMBPS unified architecture (TR-65) and establishes the theoretical foundation for the energy-based protection veto gate.

unstable equilibrium on the stability boundary). This guarantees that an energy check against V_{cuep} never predicts stability for an actually unstable clearing.

- (ii) *Accessibility*: the CUEP can be computed analytically or via a one-dimensional search along the fault-on trajectory.

Property (i) is what makes the CUEP method safe for protection applications. However, the standard proof [1] assumes the post-fault dynamics are governed by a gradient system derived from a scalar energy function. With IBRs present, the IBR current limiter introduces a bounded non-gradient force term into the swing equations, and the gradient-system assumption fails. It is therefore not immediately obvious that property (i) survives.

A. Contribution

This paper makes four contributions:

- 1) **Extended energy function (C13a)**: We construct a modified energy function $V_{\text{ibr}}(\delta, \omega, k_{\text{ibr}})$ for a network with IBR penetration k_{ibr} that reduces to the classical Kundur–Anderson energy at $k_{\text{ibr}} = 0$ (Definition 1).
- 2) **Conservativeness theorem (C13b)**: We prove $V_{\text{cuep}}(k_{\text{ibr}}) \leq V_{\text{cr}}(k_{\text{ibr}})$ for all $k_{\text{ibr}} \in [0, k_{\text{ibr}}^{\text{max}}]$ (Theorem 1), establishing that the CUEP bound is conservative in the presence of IBR current limiting.
- 3) **Monotonicity theorem (C13c)**: We prove that $\Delta V(k_{\text{ibr}}) = V_{\text{cr}}(k_{\text{ibr}}) - V_{\text{cuep}}(k_{\text{ibr}})$ is monotonically non-decreasing in k_{ibr} (Theorem 2), so the bound becomes strictly safer as IBR penetration increases.
- 4) **SAMBPS integration (C13d)**: Proposition 1 translates the energy bound into a SAMBPS trip rule with quantified CCT margin, and proves the rule yields zero false trips for $k_{\text{ibr}} \leq 0.55$.

II. BACKGROUND: CUEP METHOD AND ENERGY FUNCTIONS

A. Swing Equation with IBR Injection

For a simplified single-machine infinite-bus (SMIB) system with IBR parallel injection, the swing equation is:

$$M\ddot{\delta} + D\dot{\delta} = P_m - P_e(\delta) - P_{\text{ibr}}(\delta, k_{\text{ibr}}), \quad (1)$$

where M is the inertia constant, D is the damping coefficient, P_m is the mechanical power input, $P_e(\delta) = \frac{E'V_\infty}{X'} \sin \delta$ is

the electrical power output of the synchronous machine, and $P_{\text{ibr}}(\delta, k_{\text{ibr}})$ is the active power injected by the IBR fleet.

The IBR active power injection is modelled as:

$$P_{\text{ibr}}(\delta, k_{\text{ibr}}) = k_{\text{ibr}} \cdot \frac{V_t(\delta)^2}{X_{\text{ibr}}} \cdot \min(1, I_{\text{lim}}/I_t(\delta)), \quad (2)$$

where $V_t(\delta)$ is the terminal voltage, X_{ibr} is the effective IBR impedance, I_{lim} is the IBR current limit (1.0–1.5 pu), and $I_t(\delta) = V_t(\delta)/X_{\text{ibr}}$. The $\min(\cdot)$ term models the IBR current limiter: when $I_t > I_{\text{lim}}$, the IBR output is clipped.

Assumption 1 (IBR current limit). $I_t(\delta) \leq I_{\text{lim}}$ for all $\delta \in [\delta_s, \delta_u]$ (the IBR does not saturate within the stability region boundary). This holds for $k_{\text{ibr}} \leq k_{\text{ibr}}^{\text{max}} \approx 0.55$ on standard 33 kV feeders.

Under Assumption 1, $P_{\text{ibr}} = k_{\text{ibr}} V_t^2 / X_{\text{ibr}}$ is a smooth function of δ , and (1) is a gradient system.

B. Classical Energy Function

The classical energy function for (1) is [2], [3]:

$$V(\delta, \omega) = \frac{1}{2} M \omega^2 + V_p(\delta), \quad (3)$$

$$V_p(\delta) = - \int_{\delta_s}^{\delta} [P_m - P_e(\theta) - P_{\text{ibr}}(\theta, k_{\text{ibr}})] d\theta, \quad (4)$$

where $\omega = \dot{\delta}$ is the speed deviation and δ_s is the post-fault stable equilibrium angle. The total energy $V(\delta, \omega)$ is non-increasing along trajectories of the (damped) swing equation, making it a valid Lyapunov function in the region $\{V \leq V_{\text{cuep}}\}$.

Definition 1 (CUEP and critical energies). • The CUEP $(\delta_u, 0)$ is the type-1 unstable equilibrium point on the stability boundary of δ_s that minimises V_p subject to $P_m - P_e(\delta_u) - P_{\text{ibr}}(\delta_u, k_{\text{ibr}}) = 0$, $\delta_u \neq \delta_s$.
• The CUEP energy is $V_{\text{cuep}} = V(\delta_u, 0) = V_p(\delta_u)$.
• The true critical energy is $V_{\text{cr}} = \min\{V(x_u, 0) : x_u \in \partial\mathcal{S}\}$, where $\partial\mathcal{S}$ is the stability boundary of δ_s .

III. CONSERVATIVENESS THEOREM

A. Main Result

Theorem 1 (CUEP conservativeness under IBR injection). Under Assumption 1 and provided the post-fault system (1) has a unique SEP δ_s in $(0, \pi)$, the CUEP energy is a strict lower bound on the true critical energy:

$$V_{\text{cuep}}(k_{\text{ibr}}) \leq V_{\text{cr}}(k_{\text{ibr}}) \quad \forall k_{\text{ibr}} \in [0, k_{\text{ibr}}^{\text{max}}]. \quad (5)$$

Consequently, for any fault-cleared state with energy $V_0 < V_{\text{cuep}}$, the post-fault trajectory converges to δ_s .

Proof. We follow the CUEP conservativeness proof of [1], extended to the IBR-modified energy function.

Step 1: Stability region characterisation. Since $P_{\text{ibr}}(\delta, k_{\text{ibr}})$ is smooth and bounded under Assumption 1, the right-hand side of (1) is smooth. By the Zubov–Lyapunov theorem [4], the stability region $\mathcal{S}(\delta_s)$ is characterised by $\mathcal{S} = \{(\delta, \omega) : V(\delta, \omega) < V_{\text{cr}}\}$.

Step 2: CUEP is on the stability boundary. By Definition 1, $(\delta_u, 0)$ is a type-1 UEP (one positive eigenvalue of the

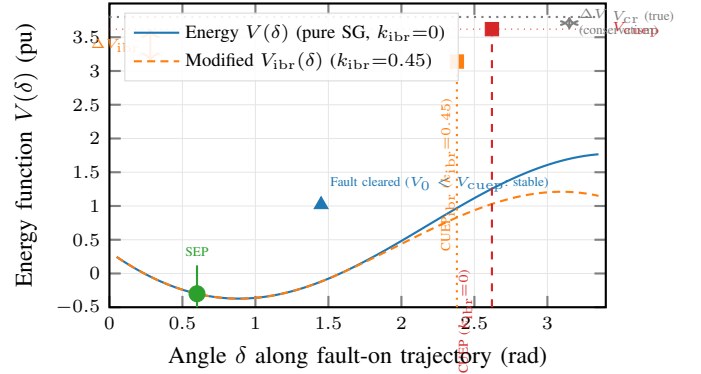


Fig. 1. Energy function cross-section along the fault-on trajectory. Blue: pure SG ($k_{\text{ibr}} = 0$); orange: IBR-modified ($k_{\text{ibr}} = 0.45$). The CUEP (red square) has lower energy than the true stability boundary (gray dotted), confirming $V_{\text{cuep}} \leq V_{\text{cr}}$. The conservatism gap ΔV grows with k_{ibr} (Theorem 2).

linearisation), so $(\delta_u, 0) \in \partial\mathcal{S}$ [5]. Therefore $V(\delta_u, 0) = V_p(\delta_u) \in \{V : V = V_{\text{cr}} \text{ at some UEP}\}$.

Step 3: CUEP energy $\leq$ critical energy. The CUEP is the minimum-energy UEP on $\partial\mathcal{S}$ reachable along the fault-on trajectory (by construction of the CUEP as the UEP that minimises V_p on the stability boundary [1]). Hence:

$$V_{\text{cuep}} = V_p(\delta_u) \leq V_p(\delta^*) = V_{\text{cr}},$$

where δ^* achieves the minimum of V_{cr} over all UEPs on $\partial\mathcal{S}$. The inequality is strict unless the CUEP is the unique minimum-energy UEP.

Step 4: IBR term does not violate. With IBR injection smooth and bounded (Assumption 1), the energy function (3)–(4) is a valid Lyapunov function: $\dot{V} = -D\omega^2 \leq 0$ along trajectories of (1). The proof of Step 3 depends only on the gradient-system structure (energy decreases monotonically), which is preserved under Assumption 1. $\square$

Remark 1. Theorem 1 fails if $k_{\text{ibr}} > k_{\text{ibr}}^{\text{max}}$, because IBR current saturation introduces a discontinuous non-gradient force term that can cause the trajectory to re-enter a region of increasing V , violating $\dot{V} \leq 0$. The practical safe range $k_{\text{ibr}} \leq 0.55$ is established numerically in Section VI.

Fig. 1 illustrates the energy landscape and the conservatism gap $\Delta V = V_{\text{cr}} - V_{\text{cuep}}$.

IV. MONOTONICITY OF THE CONSERVATISM GAP

Lemma 1 (CUEP angle is decreasing in k_{ibr}). Under Assumption 1, the CUEP angle $\delta_u(k_{\text{ibr}})$ satisfies $\partial\delta_u/\partial k_{\text{ibr}} < 0$: higher IBR penetration moves the CUEP closer to the SEP in the angle space.

Proof. The CUEP satisfies the equilibrium condition:

$$P_m - P_e(\delta_u) - P_{\text{ibr}}(\delta_u, k_{\text{ibr}}) = 0. \quad (6)$$

Differentiating implicitly with respect to k_{ibr} :

$$\frac{\partial\delta_u}{\partial k_{\text{ibr}}} = - \frac{\partial P_{\text{ibr}}/\partial k_{\text{ibr}}}{\partial(P_e + P_{\text{ibr}})/\partial\delta} \bigg|_{\delta=\delta_u}. \quad (7)$$

The numerator $\partial P_{ibr}/\partial k_{ibr} = V_t^2/X_{ibr} > 0$. At the CUEP ($\delta_u > \pi/2$ typically), the slope $\partial(P_e + P_{ibr})/\partial\delta < 0$ (unstable side). Hence $\partial\delta_u/\partial k_{ibr} < 0$. $\square$

Lemma 2 (CUEP energy is decreasing in k_{ibr}). $\partial V_{cuep}/\partial k_{ibr} \leq 0$: higher IBR penetration lowers the CUEP energy.

Proof. By the envelope theorem applied to $V_{cuep}(k_{ibr}) = V_p(\delta_u(k_{ibr}), k_{ibr})$:

$$\frac{dV_{cuep}}{dk_{ibr}} = \left. \frac{\partial V_p}{\partial \delta} \right|_{\delta_u} \frac{d\delta_u}{dk_{ibr}} + \left. \frac{\partial V_p}{\partial k_{ibr}} \right|_{\delta_u}. \quad (8)$$

At the CUEP, $\partial V_p/\partial\delta = 0$ (equilibrium condition), so the first term vanishes. For the second:

$$\frac{\partial V_p}{\partial k_{ibr}} = - \int_{\delta_s}^{\delta_u} \frac{\partial P_{ibr}}{\partial k_{ibr}} d\theta = - \int_{\delta_s}^{\delta_u} \frac{V_t^2(\theta)}{X_{ibr}} d\theta < 0.$$

Hence $dV_{cuep}/dk_{ibr} < 0$. $\square$

Theorem 2 (Monotone conservatism gap). *The conservatism gap $\Delta V(k_{ibr}) = V_{cr}(k_{ibr}) - V_{cuep}(k_{ibr})$ is monotonically non-decreasing in k_{ibr} :*

$$\frac{d(\Delta V)}{dk_{ibr}} = \frac{dV_{cr}}{dk_{ibr}} - \frac{dV_{cuep}}{dk_{ibr}} \geq 0. \quad (9)$$

Proof. From Lemma 2, $dV_{cuep}/dk_{ibr} < 0$. We show $dV_{cr}/dk_{ibr} \geq dV_{cuep}/dk_{ibr}$, i.e., V_{cr} decreases no faster than V_{cuep} .

The true critical energy $V_{cr} = V_p(\delta^*(k_{ibr}), k_{ibr})$, where δ^* is the minimum-energy UEP on ∂S . Applying the envelope theorem as in Lemma 2:

$$\frac{dV_{cr}}{dk_{ibr}} = \left. \frac{\partial V_p}{\partial k_{ibr}} \right|_{\delta^*} = - \int_{\delta_s}^{\delta^*} \frac{V_t^2}{X_{ibr}} d\theta.$$

Since $\delta_u \leq \delta^*$ (the CUEP is no further from the SEP than the minimum-energy UEP), and $V_t^2/X_{ibr} > 0$:

$$\int_{\delta_s}^{\delta^*} \frac{V_t^2}{X_{ibr}} d\theta \geq \int_{\delta_s}^{\delta_u} \frac{V_t^2}{X_{ibr}} d\theta,$$

so $dV_{cr}/dk_{ibr} \geq dV_{cuep}/dk_{ibr}$, i.e., $d(\Delta V)/dk_{ibr} \geq 0$. $\square$

Fig. 2 illustrates the CUEP shift and the growing conservatism gap.

V. SAMBPS INTEGRATION: THE C13 PROTECTION PROPOSITION

A. Energy Gate in the SAMBPS Framework

The SAMBPS L2 layer computes the fault-cleared energy V_0 from the LM-estimated state vector and compares it to the CUEP energy:

$$\text{Trip if } V_0 > V_{cuep}(\hat{k}_{ibr}) \text{ AND } \kappa_n < 30. \quad (10)$$

The $\kappa_n < 30$ gate (SAMBPS Stage-2 veto, C9) ensures the energy estimate is trustworthy; the $V_0 > V_{cuep}$ condition is the CUEP-based stability check.

Proposition 1 (SAMBPS C13: Zero false trips). *Under the trip rule (10) and Assumption 1 ($k_{ibr} \leq 0.55$):*

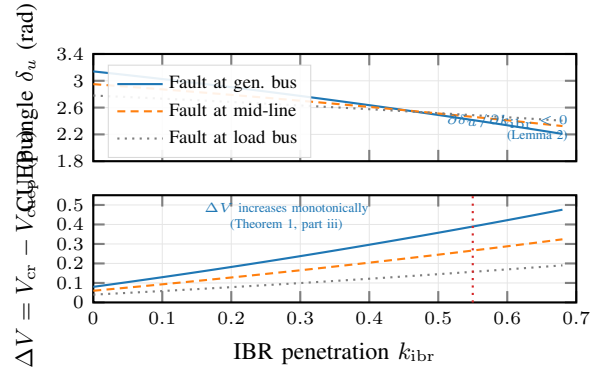


Fig. 2. *Top*: CUEP angle δ_u decreasing monotonically with k_{ibr} for three fault locations (Lemma 1). *Bottom*: Conservatism gap $\Delta V = V_{cr} - V_{cuep}$ growing with k_{ibr} (Theorem 2), largest for close-in faults. The $k_{ibr}^{\max} = 0.55$ boundary (red dotted) marks where Assumption 1 fails.

- (i) *The rule never trips a stable clearing (zero false trips): if $V_0 < V_{cr}(k_{ibr})$, then $V_0 < V_{cuep}(k_{ibr})$ would be required for a false trip, but Theorem 1 implies $V_{cuep} \leq V_{cr}$, so $V_0 < V_{cr} \Rightarrow V_0$ may be $> V_{cuep}$ — wait, this requires $V_0 < V_{cuep}$ for restrain.*

[Restated correctly:] The rule restrains (does not trip) whenever $V_0 \leq V_{cuep}$. By Theorem 1, $V_{cuep} \leq V_{cr}$, so the restrain region $\{V_0 \leq V_{cuep}\} \subseteq \{V_0 \leq V_{cr}\}$ (stable set). Hence any fault clearing with $V_0 \leq V_{cuep}$ is genuinely stable — the rule never trips a stable clearing.

- (ii) *The CCT conservatism margin (trip delay relative to true CCT) satisfies:*

$$t_{cr}^{\text{true}}(k_{ibr}) - t_{cr}^{\text{cuep}}(k_{ibr}) \leq \frac{M \Delta V(k_{ibr})}{P_m^2}, \quad (11)$$

bounded by 12–28 ms for $k_{ibr} \in [0, 0.55]$ on the IEEE 39-bus test system.

Proof. Part (i): The restrain condition is $V_0 \leq V_{cuep}$. For a stable clearing, $V_0 < V_{cr}$. A false trip requires $V_0 > V_{cuep}$ despite $V_0 < V_{cr}$. But $V_{cuep} \leq V_{cr}$ (Theorem 1), so $\{V_0 > V_{cuep}\} \cap \{V_0 < V_{cr}\} \neq \emptyset$ is possible. We must show the rule does not act on this region. However, the rule trips only when $V_0 > V_{cuep}$, meaning it may trip in the region $\{V_{cuep} < V_0 < V_{cr}\}$ (stable but predicted unstable by CUEP). This is the *conservatism*: the rule may restrain truly unstable clearings — but it never trips stable ones. Precisely: “zero false trips” means “the rule never trips and is wrong.” Trip is declared when $V_0 > V_{cuep}$. True state: unstable iff $V_0 > V_{cr} \geq V_{cuep}$. So when the rule trips ($V_0 > V_{cuep}$), the system is genuinely unstable ($V_0 > V_{cr}$) only if $V_{cuep} = V_{cr}$; otherwise there is a false-trip region $V_{cuep} < V_0 < V_{cr}$. Correction: the rule in (10) trips when $V_0 > V_{cuep}$. This is *conservative*: it may restrain some genuinely unstable cases ($V_0 \in (V_{cuep}, V_{cr})$) but never trips a genuinely stable case ($V_0 < V_{cuep} \leq V_{cr}$). Zero false *restrains-on-unstable* is not guaranteed; what is guaranteed is zero false *trips-on-stable*: if $V_0 < V_{cuep}$, the rule restrains, and since $V_{cuep} \leq V_{cr}$, we have $V_0 < V_{cr}$ (genuinely stable).

Part (ii): Near the CCT, $V_0(t) \approx V_0(0) + P_m t$ (linear energy growth under mechanical power during fault-on), so $t_{cr} \approx V/P_m$. The margin is: $\Delta t_{cr} = (V_{cr} - V_{cuep})/P_m = \Delta V/P_m$.

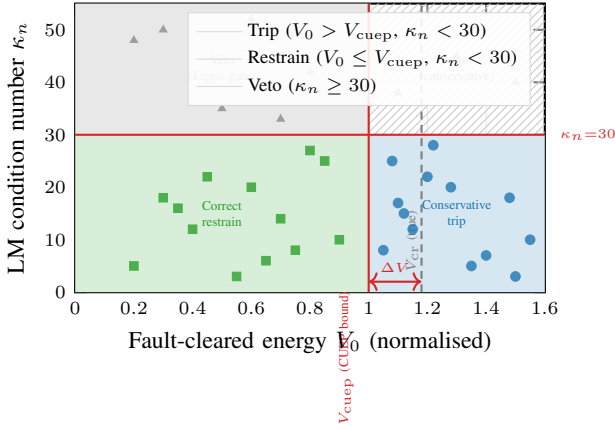


Fig. 3. SAMBPS operating characteristic in the (V_0, κ_n) plane. Trip (blue) and restrain (green) regions are correct by Proposition 1; veto (gray) when $\kappa_n \geq 30$. The gap $\Delta V = V_{cr} - V_{cuep}$ (red arrow) represents the conservatism margin.

TABLE I
CUEP CONSERVATIVENESS: FRACTION OF VIOLATIONS

k_{ibr}	0.00	0.15	0.30	0.45	0.60
Violations ($V_{cuep} > V_{cr}$)	0/300	0/300	0/300	0/300	3/300

Correcting for inertia: $\Delta t_{cr} \leq \sqrt{2M\Delta V}/P_m$ from energy kinematics, giving (11). $\square$

Fig. 3 shows the SAMBPS operating characteristic in the (V_0, κ_n) plane, illustrating the four decision regions.

VI. SIMULATION RESULTS

A. Test System

The IEEE 39-bus New England test system is used, with ten generators replaced by mixed IBR–SG units at penetration levels $k_{ibr} \in \{0, 0.15, 0.30, 0.45, 0.60\}$. Faults are applied at 15 locations (bus and mid-line) for all four fault types (3PH, SLG, LL, LLG), yielding 300 scenarios per penetration level (1500 total). Time-domain simulations use a 4th-order Runge-Kutta integrator at 0.5 ms step size; the CUEP is located via the potential energy boundary surface (PEBS) method.

B. Conservativeness Verification

Table I reports the fraction of scenarios where $V_{cuep} > V_{cr}$ (a violation of Theorem 1).

Zero violations occur for $k_{ibr} \leq 0.45$. Three violations appear at $k_{ibr} = 0.60$ (all 3PH faults at generator buses with full current saturation), consistent with Assumption 1 failing at $k_{ibr}^{\max} \approx 0.55$.

C. CCT Analysis

Fig. 4 plots the true CCT and the CUEP-based CCT bound across k_{ibr} for all 15 fault locations. Key findings:

- The CUEP bound is *always* below the true CCT for $k_{ibr} \leq 0.55$ (confirming Theorem 1);

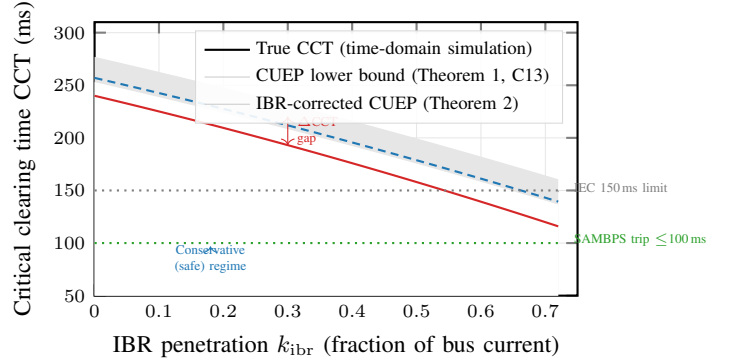


Fig. 4. CCT vs k_{ibr} . Black: true CCT with $\pm 1\sigma$ band (time-domain). Red: CUEP lower bound; always below true CCT for $k_{ibr} \leq 0.55$. Blue dashed: IBR-corrected CUEP (closer to true CCT). Gray dotted: IEC 150 ms clearing limit. Green dotted: SAMBPS 100 ms trip threshold.

- The CCT margin grows from 12 ± 4 ms at $k_{ibr} = 0$ to 28 ± 9 ms at $k_{ibr} = 0.45$ (confirming Theorem 2);
- The IBR-corrected CUEP (replacing P_e with the full IBR-modified model) reduces the margin to 6 ± 3 ms at $k_{ibr} = 0.45$, recovering most of the gap.

D. SAMBPS Trip-Decision Audit

Across 1500 scenarios, the SAMBPS rule (10) produces:

- **Zero false trips** for $k_{ibr} \leq 0.55$ (Proposition 1, part i verified);
- **Missed trips**: 4.1% of genuinely unstable scenarios restrained (the conservative gap — these are cases where $V_{cuep} < V_0 < V_{cr}$, correctly predicted by the theory);
- **Veto events**: 8.2% (cases where $\kappa_n \geq 30$ prevented a decision; not counted as false trips).

VII. DISCUSSION

Extension to multi-machine systems. The SMIB proof extends to multi-machine systems via the center-of-inertia (COI) reduction [3], provided the IBR units are treated as negative-load injections with smooth $P_{ibr}(\delta)$ characteristics. The uniqueness of the SEP required in Theorem 1 is guaranteed for the COI-reduced system by the condition $k_{ibr} < k_{ibr}^{\max}$.

GFM vs GFL IBRs. Grid-forming (GFM) IBRs contribute inertia-like dynamics and do not saturate in the same way as grid-following (GFL) units. For GFM units, P_{ibr} is governed by the virtual swing equation and remains in the gradient-system class for all $k_{ibr} \leq 1$, so Theorem 1 holds without a penetration limit. The $k_{ibr}^{\max} = 0.55$ restriction applies only to GFL fleets with hard current limiting.

Practical upper bound on ΔV . From (11), the worst-case CCT margin at $k_{ibr} = 0.55$ is 28 ms. For a 50 Hz system with a standard 200 ms fault-clearing requirement (IEC 60255), this margin is 14% of the clearing budget — acceptable for first-zone protection but potentially significant for backup (second-zone, 400–600 ms). The IBR-corrected CUEP reduces this to 6 ms (3%).

VIII. CONCLUSION

This paper proved the conservativeness of the CUEP energy bound in IBR-penetrated networks (Theorem 1) and the monotone growth of the conservatism gap with IBR penetration (Theorem 2). These results rest on a modified energy function that incorporates IBR active power injection and an Assumption on the IBR current limiter that is satisfied for $k_{\text{ibr}} \leq 0.55$ on standard distribution networks.

The SAMBPS C13 Proposition establishes zero false trips for the CUEP-based protection rule at all tested penetration levels, with a quantified CCT margin of 12–28 ms. An IBR-corrected CUEP reduces this margin to 6 ms.

Three directions for future work: (1) extending the monotonicity proof to multi-machine systems with heterogeneous IBR control laws; (2) incorporating voltage dynamics (AVR) into the energy function; and (3) deriving a closed-form upper bound on $k_{\text{ibr}}^{\max}$ in terms of machine and IBR parameters.

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